



Cite this: *Org. Biomol. Chem.*, 2014, **12**, 7900

Received 7th July 2014,
Accepted 26th August 2014

DOI: 10.1039/c4ob01405g

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Zinc mediated activation of terminal alkynes: stereoselective synthesis of alkynyl glycosides†‡

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Zinc mediated alkynylation reaction was studied for the preparation of C-glycosides from unactivated alkynes. Different glycosyl donors such as glycals and anomeric acetates were tested towards an alkynyl zinc reagent obtained from alkynes using zinc dust and ethyl bromoacetate as an additive. The method provides simple, mild and stereoselective access to alkynyl glycosides both from aromatic and aliphatic acetylenes.

The synthesis of chiral building blocks from cheap and easily available starting materials has always been an interest in organic chemistry.¹ In this endeavour, carbohydrates have uninterruptedly maintained a prominent place. Though these naturally occurring materials on their own serve to be starting materials for chiral synthesis,² attachment of extra carbon chains gives rise to new chiral scaffolds of synthetic and biological importance.³ In this regard, alkynyl C-glycosides have attracted tremendous interest due to the presence of multiple bonds; a real asset for target oriented synthesis and creation of diversity.⁴ With our continuing interest in this direction,⁵ we have reported direct access to alkynyl C-glycosides from unactivated aromatic alkynes under copper triflate/ascorbic acid catalysis.^{5a} Despite several advantages over the literature methods,⁶ the main demerit of our reagent system lay in its inefficiency to activate aliphatic alkynes and glycosyl donors, except for glycals. In order to overcome these shortcomings of our strategy, we continued our studies in developing a universal reagent for C-alkynylation of glycals and other carbohydrate donors.

C–C bond formation reactions using *in situ* generated organo-zinc species have thoroughly been investigated in unsaturated systems.⁷ Low reactivity and broad functional group

compatibilities of these organo-metallic species have widened their applications in synthetic organic chemistry.⁷ In carbohydrate chemistry, however, the use of organo-zinc reagents is limited due to solubility problems, such as the use of a combination of solvents,⁸ use of strong base,⁹ poor anomeric selectivity,¹⁰ oxidation–reduction sequence,¹¹ excess use of promoters and drastic reaction conditions.⁸ To the best of our understanding, there are no reports of the applications of organo-zinc reagents for C-alkynylation of glycals and other glycosyl donors. In this letter, we would like to disclose our findings regarding C-alkynylation in carbohydrates with high stereocontrol obviating the use of strong Lewis acids or bases and exploiting both aromatic and aliphatic alkynes.

In order to develop a reagent system for C-alkynylation of glycals from unactivated aliphatic alkynes, 3,4,6-tri-O-acetyl-D-glucal (**1**) and 1-octyne (**2**) were allowed to react in the presence of various Lewis acids and metals; the results are summarized in Table 1. Initially we attempted to use Lewis acids, like Zn & In halides (Table 1, entries 1–3), where we observed the formation of alkynyl glycoside **3** and halogenated vinyl glycoside **3'** in equal amounts. Changing the metal halide with iron chloride failed to give the desired product **3**, instead we obtained complex mixtures of halogenated vinyl glycoside **3'** (Table 1, entry 4).

Triflates such as Fe(OTf)₃, In(OTf)₃, TMSOTf (Table 1, entries 5–7) exclusively furnished the desired product **3**, however, in poor yields (18–20%). Taking a clue from the literature reports for zinc mediated alkyne activation,¹² when the above reaction was carried out using zinc powder in the absence of any additive, the starting material was recovered as such (Table 1, entry 8). We observed that the choice of additive and the nature of the solvent play a significant role in the outcome of the reaction. After extensive investigation of additives and solvents, the formation of the desired C-alkynyl glycoside (**3**) (68%) was witnessed with zinc powder in the presence of ethyl bromoacetate as an additive in DCM at 40 °C (Table 1, entry 9). Other solvents like THF (Table 1, entry 10) or other additives like methyl iodide (Table 1, entry 12) failed to give the desired product. Replacing zinc with indium produced the desired product in lower yield (Table 1, entry 11).

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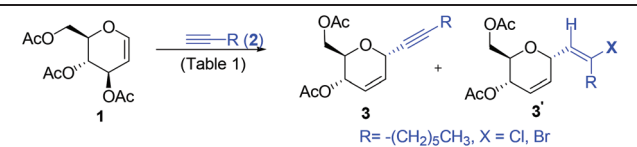
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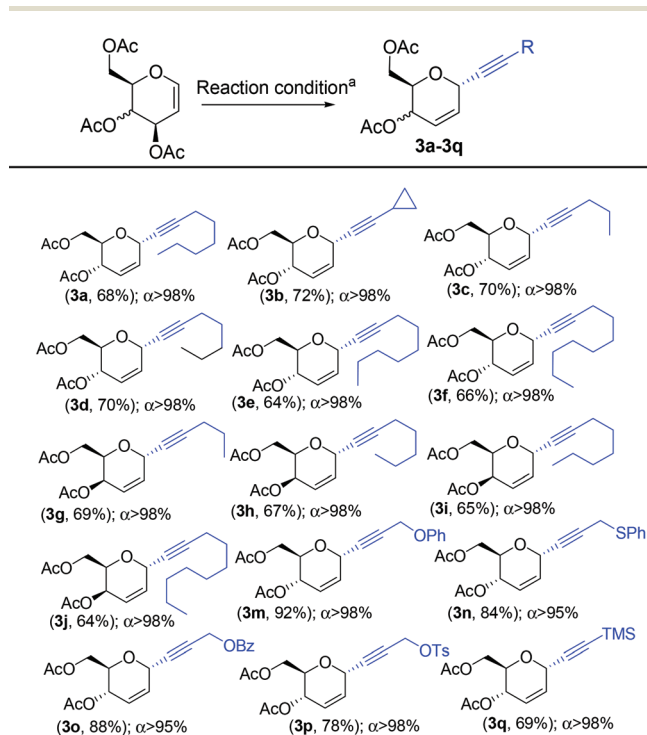
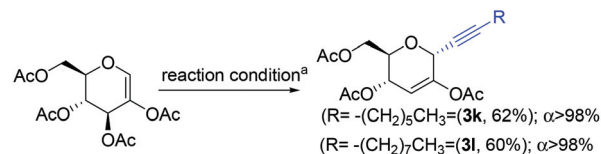
‡ Electronic supplementary information (ESI) available: Detailed experimental procedures and compound characterization data is given. See DOI: 10.1039/c4ob01405g

Table 1 Standardization of reaction conditions for C-alkynylation


Entry	Reagent ^a	Solvent/T (°C)	Time (h)	Yield ^b (%) (3 : 3')
1	ZnBr ₂ (0.2 eq.)	DCM/rt to 40	5	72 (1 : 1)
2	ZnCl ₂ (0.2 eq.)	DCM/rt to 40	5	70 (1 : 1)
3	InCl ₃ (0.2 eq.)	DCM/rt to 40	3	40 (1 : 1)
4	FeCl ₃ (0.2 eq.)	DCM/rt to 40	6	46 (0 : 1) ^c
5	Fe(OTf) ₃ (0.2 eq.)	DCE/0 to RT	16	20 (1 : 0)
6	In(OTf) ₃ (0.2 eq.)	DCE/0 to RT	16	20 (1 : 0)
7	TMSOTf (0.2 eq.)	DCE/0 to RT	16	18 (1 : 0)
8	Zn (1.5 eq.)	DCM/40	24	NR
9	Zn (1.5 eq.)/A ^d	DCM/40	6	68 (1 : 0)
10	Zn (1.5 eq.)/A	THF/40	12	NR
11	In (1.5 eq.)/A	DCM/40	12	34 (1 : 0)
12	Zn (1.5 eq.)/B ^e	DCM/40	12	NR

^a In all the cases 0.5 mmol of **1**, 0.75 mmol of (1-octyne) **2**. ^b Combined yield of isolated products (**3**, **3'**), obtained after column chromatography. ^c A combined yield of complex mixture. ^d A = BrCH₂COOC₂H₅ (1 eq.). ^e B = CH₃I (1 eq.).

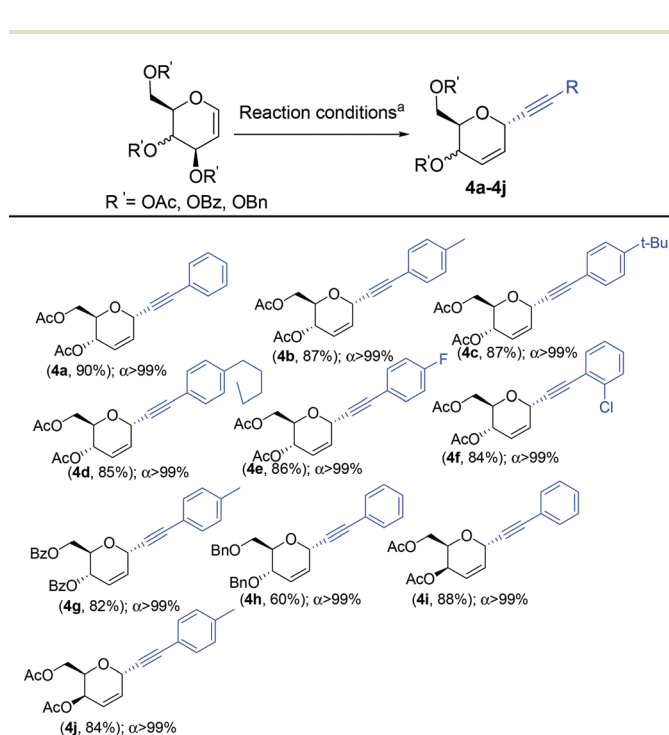
With the optimized reaction conditions at our disposal, a series of aliphatic C-alkynes with different chain lengths and additional functionalities were allowed to react with glycals and the results are summarized in Scheme 1. All the alkynes

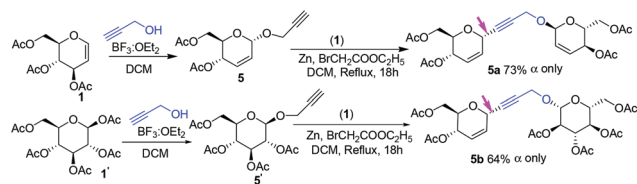
**Scheme 1** Scope of reaction with aliphatic alkynes. ^a Conditions: bromoethylacetate (1.0 eq.), alkyne (1.5 eq.), zinc dust (1.5 eq.), CH₂Cl₂ (3 mL), 40 °C, 1 h, then glycal (1.0 eq.), 40 °C, 6–8 h.**Scheme 2** Reaction scope with 2-acetoxyglucal.

smoothly reacted to yield the desired product in moderate to good yield with high stereoselectivity.

Glucal (**3a–3f**), galactal (**3g–3j**) substrates (Scheme 2) afforded C-glycosides with high selectivity. In addition to these, reaction of terminal alkynes containing *O*- and *S*-substituents (**3m**, **3n**), including ester (**3o**) and tosylate (**3p**) yielded the desired product, expanding the scope of the reacting partners. One of the key features of this methodology is its mildness as evident from the reaction of TMS acetylene under similar conditions leading to the formation of alkynylated glycoside (**3q**), retaining the silyl protection at the terminal position which is otherwise not possible.

Similarly, in the case of 2-acetoxy D-glucals, we obtained 2-acetoxy glycosides containing enol ester moiety, whereas 2-keto glycosides are the major product with other Lewis acids under harsh conditions.^{4a} After getting encouraging results with aliphatic acetylenes, we tried to make this protocol more general by studying its applications for aromatic alkynes. As anticipated, aromatic alkynes gave considerable higher yields as compared to the aliphatic ones and the results are depicted in Scheme 3. Glycals bearing ester (**4a–4j**) and ether (**4h**) pro-

**Scheme 3** Substrate scope of reaction towards aromatic alkynes. ^a Bromoethylacetate (1.0 equiv.), alkyne (1.5 equiv.), zinc dust (1.5 equiv.), CH₂Cl₂ (3 mL), 40 °C, 1 h, then glycal (1.0 equiv.), 40 °C, 4–5 h.

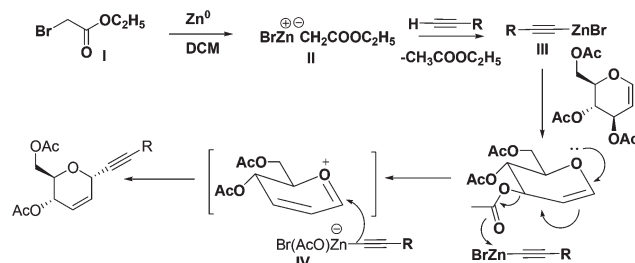


Scheme 4 Synthesis of pseudo-disaccharides.

protecting groups smoothly reacted with a range of aryl alkynes carrying electron donating or withdrawing groups to give the corresponding alkynyl *C*-pseudo glycals in good to excellent yields comparable with the TMSOTf conditions or superior to the copper triflates/ascorbic acid conditions.^{5a}

To demonstrate the versatility of the reagent system, pseudo-disaccharides **5a**, **5b** were prepared (Scheme 4). Thus, treatment of propargyloxy-glycoside **5**, **5'** derived from D-glucal (**1**) or glucose penta-acetate (**1'**) with tri-*O*-acetyl-D-glucal under optimized conditions led to the formation of disaccharides in moderate yields.

In a further extension of this strategy, other glycosyl donors like anomeric acetates were allowed to react with alkynes



Scheme 5 Plausible mechanism of C-alkynylation.

under optimized reaction conditions. It was observed that all the donors reacted with both aliphatic and aromatic alkyne partners to afford the corresponding glycosides (**6a**, **6b**; **7a**, **7b**) in good to excellent yield with high stereoselectivity, and the results are represented in Table 2. However, in this case of per-acetylated glucose, we failed to observe any *C*-glycosylated product even after 48 h.

In order to explain the alkyne activation under the standardized conditions, a plausible reaction mechanism is outlined in Scheme 5. Formation of an organo-zinc compound (**II**) from α-halo ester (**I**) and zinc metal is well known.^{7c,12} Exchange of an sp-proton between the terminal alkynes and organo-zinc species (**II**) leads to the formation of zinc acetylide (**III**) with the removal of ethyl acetate. Subsequently, the zinc acetylide reacts with glycal *via* an allylic oxocarbenium intermediate and a zincate complex (**IV**) at C-1 from the α-side, to afford the desired *C*-glycoside with α-selectivity. Isobe^{6d,13} explained the α-selectivity in the case of alkynylation with trimethyl acetylenes by electronic effects on the oxocarbenium intermediates involved in the transformation. In the case of anomeric acetates, the reaction mechanism proceeds through an oxocarbenium intermediate, resulting from the expulsion of anomeric acetate.^{6e}

In conclusion, we have developed a new and very mild zinc-mediated method for the stereoselective introduction of an unactivated alkynyl group to glycals and glycosyl acetates. This offers convenient access to obtain *C*-glycosides with good yields and it has a broad substrate scope.

Table 2 Direct C-glycosylation of unactivated alkynes with glycosyl acetates

Entry	Glycosyl acetate	Product	Yield (%)
1			64
2			60
3			60
4			58

^a Bromoethylacetate (1.0 equiv.), alkyne (1.5 equiv.), zinc dust (1.5 equiv.), CH₂Cl₂ (3 mL), 40 °C, 1–2 h, then glycosyl acetate (1.0 equiv.), 40 °C, 24 h.

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